

Chapter 3 Lab: Designing and Securing an Azure Virtual Network

Task 1: Create a Virtual Network (VNet)

The screenshot displays the 'Create virtual network' wizard in the Microsoft Azure portal. The URL in the browser is <https://portal.azure.com/#create/Microsoft.VirtualNetwork-ARM>. The page has a blue header with the Microsoft Azure logo and a search bar. The main content area is titled 'Create virtual network' and includes tabs for 'Basics', 'Security', 'IP addresses', 'Tags', and 'Review + create'. The 'Basics' tab is active. Below the tabs, there is a section for 'Project details' with a description: 'Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.' The 'Subscription' dropdown is set to 'Azure subscription 1' and the 'Resource group' dropdown is set to '(New) RG-NetworkLab'. There is a 'Create new' link below the resource group dropdown. Below this is the 'Instance details' section with the 'Virtual network name' set to 'VNet-Company' and the 'Region' set to '(Africa) South Africa North'. There is a 'Deploy to an Azure Extended Zone' link below the region dropdown. At the bottom of the wizard, there are three buttons: 'Previous', 'Next', and 'Review + create' (which is highlighted in blue). A 'Give feedback' link is also present in the bottom right corner.

I created a new resource group called **“RG-NetworkLab”** using **one Azure subscription** to manage and organize all resources within the environment. This resource group acts as a container for related resources, simplifying management, monitoring, and access control, while allowing easy deployment or deletion of all associated resources. Within this group, I deployed a **Virtual Network (VNet)**, which serves as the foundational building block for networking in Azure. The VNet enables virtual machines and services to communicate securely with each other, connect to the internet, and integrate with on-premises networks, while providing isolation, scalability, and segmentation similar to a traditional data center network. The region selected was **South Africa North**, as it is geographically close, reducing latency, ensuring compliance, and optimizing performance and cost. This setup provides a secure, organized, and scalable network foundation for all lab resources.

Microsoft Azure

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Create virtual network

BasicsSecurityIP addressesTagsReview + create

Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you need. [Learn more](#)

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. [Learn more](#)

☐ Allocate using IP address pools. [Learn more](#)

+ Add a subnet

10.0.0.0/16

Delete address space

This address prefix overlaps with virtual network 'VM-Linux-Nilbo-vnet'. If you intend to peer these virtual networks, change the address space. [Learn more](#)

10.0.0.0

/16

10.0.0.0 - 10.0.255.25565,536 addresses

https://portal.azure.com/#create/Microsoft.VirtualNetwork-ARM

Microsoft Azure

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Create virtual network

BasicsSecurityIP addressesTagsReview + create

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10.0.0.0

/16

10.0.0.0 - 10.0.255.25565,536 addresses

Subnets	IP address range	Size	NAT gateway
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)	-
WebSubnet	10.0.1.0 - 10.0.1.255	/24 (256 addresses)	-
DBSubnet	10.0.2.0 - 10.0.2.255	/24 (256 addresses)	-

Previous

Next

Review + create

Give feedback

← ↻ https://portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/_/overview/id/%2Fsubscriptions%2Fbfe2eef6-3077-4a69-... Microsoft Azure Search resources, services, and docs (G+J) Copilot kperalr1bo@gmail.com

Home VNet-Company-1775142187328 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview Inputs Outputs Template

✓ Your deployment is complete

Deployment name : VNet-Company-1775142187328 Start time : 4/2/2026, 5:03:14 PM
Subscription : Azure subscription 1 Correlation ID : cff384a1-605d-41ef-8276-3eaaded61f4a
Resource group : RG-NetworkLab

Deployment details

Resource	Type	Status	Operation details
VNet-Company	Virtual network	OK	Operation details

Next steps

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

Microsoft Defender for Cloud
Secure your apps and infrastructure
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Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
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```
Value[0]      :  
  Code       : ComponentStatus/StdOut/succeeded  
  Level      : Info  
  DisplayStatus : Provisioning succeeded  
  Message    : Success Restart Needed Exit Code      Feature Result  
-----  
True   No           Success      {Common HTTP Features, Default Document, D...  
  
PS /home/luyanda>  
PS /home/luyanda> ComponentStatus/StdErr/succeeded  
  Level      : Info  
  DisplayStatus : Provisioning succeeded  
  Message    :  
Status      : Succeeded  
Capacity    : 0  
Count       : 0
```

```

Web-Server -IncludeManagementTools"

value[0]
:
Code      : ComponentStatus/StdOut/succeeded
Level     : Info
DisplayStatus : Provisioning succeeded
Message    : Success Restart Needed Exit Code      Feature Result
-----
true      No          Success          {Common HTTP Features, Default Document, D...

value[1]
:
Code      : ComponentStatus/StdErr/succeeded
Level     : Info
DisplayStatus : Provisioning succeeded
Message    :
Status    : Succeeded
Capacity  : 0
Count     : 0

```

Task 2: Deploy Two Web Servers in Web Subnet

Deploying two web servers in the same Web Subnet in Azure is a best practice for building resilient, scalable, and highly available applications.

The screenshot displays the 'Create a virtual machine' wizard in the Microsoft Azure portal. The configuration is as follows:

- Resource group:** RG-NetworkLab
- Virtual machine name:** VM-Web1
- Region:** (Africa) South Africa North
- Availability options:** Availability zone
- Zone options:** Self-selected zone (selected)
- Availability zone:** Zone 1
- Security type:** Trusted launch virtual machines
- Image:** Windows Server 2022 Datacenter: Azure Edition Hotpatch - x64 Gen2

Estimated monthly costs:

Category	Item	Cost
Basics	Virtual machine	\$159.87
	Image	\$67.16
	Size	\$92.71
Disks		\$0.00
Total	Estimated monthly cost	\$163.52

← ↻ <https://portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2Fbfe2eef6-3077-4a69-...> Copilot kporalntibo@gmail.com

Home

CreateVm-microsoftwindowsserver.windowsserver2022-20260402173107 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

✓ Your deployment is complete

Deployment name : CreateVm-microsoftwindowsserver.windowsserver2022-20260402... Start time : 4/2/2026, 5:37:41 PM
Subscription : Azure subscription 1 Correlation ID : 3c349a54-2a51-4784-9f86-5d4815ee3a13
Resource group : RG-NetworkLab

Deployment details

Resource	Type	Status	Operation details
VM-Web1	Virtual machine	OK	Operation details
vm-web1596_z1	Microsoft.Network/networkInter	OK	Operation details
VM-Web1-nsg	Network security group	OK	Operation details
VM-Web1-ip	Public IP address	OK	Operation details

Next steps

- Set up auto-shutdown Recommended
- Monitor VM health, performance, and network dependencies Recommended
- Run a script inside the virtual machine Recommended

[Go to resource](#) [Create another VM](#) [Scale out your VM](#)

Give feedback
Add or remove favorites by pressing Ctrl+Shift+F
[Tell us about your experience with deployment](#)

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

← ↻ <https://portal.azure.com/#create/Microsoft.VirtualMachine> Copilot kporalntibo@gmail.com

Home

Create a virtual machine

Subscription * Azure subscription 1
Resource group * RG-NetworkLab
[Create new](#)

Instance details

Virtual machine name * VM-Web2 ✓
Region * (Africa) South Africa North
[Deploy to an Azure Extended Zone](#)
Availability options Availability zone
Zone options ☒ Self-selected zone
Choose up to 3 availability zones, one VM per zone
☐ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs
Availability zone * Zone 1
☒ You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type Trusted launch virtual machines
[Configure security features](#)

Image * Windows Server 2022 Datacenter: Azure Edition Hotpatch - x64 Gen2

< Previous Next: Disks > [Review + create](#)

Estimated monthly costs

Costs indicated here are estimates only. Pricing may vary depending on your Microsoft agreement, date of purchase, subscription type, usage costs, licensing and currency exchange rates. Total costs may include other resource costs, licensing and subscription implications. This feature may have limited or restricted functionality, but is made available on a preview basis for evaluation and feedback.

[Give feedback about your estimate experience](#)

Basics		\$159.87
Virtual machine		\$159.87
Image	Windows Server 2022 Datacenter: Azure Edition Hotpatch	\$67.16
Size	Standard_D2s_v3	\$92.71
> Disks		\$0.00
> Networking		\$0.00
Estimated monthly cost		\$159.87

[Give feedback](#)

Create two Virtual Machines, named VM-Web1 and VM-Web2, with both using the same image and size. Image = Windows Servers 2025 Datacenter: Azure Edition – x64 Gen2 Size = Standard_B2ats_v2 – 2 vcpus Windows Server 2025 Datacenter: Azure Edition – x64 Gen2 with Standard_B2ats_v2 (2 vCPUs) in Azure is about balancing modern OS features, Azure integration, and cost-effective compute resources. Standard_B2ats_v2 (2 vCPUs) in Azure

is suitable for lightweight to medium workloads such as web servers, dev/test environments, or small business apps.

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network *	VNet-Company Create new
Subnet *	Subnet2 (10.0.2.0/24) Manage subnet configuration
Public IP	(new) VM-Web2-ip Create new Public IP addresses have a nominal charge. Estimate price
NIC network security group	☐ None ☒ Basic ☐ Advanced
Public inbound ports *	☐ None

Home > VM-Web1

VM-Web1 | Connect

Virtual machine

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Monitor
- Resource visualizer
- Connect
 - Connect
 - Bastion
 - Windows Admin Center
- Networking
- Settings
- Availability + scale
- Security
- Backup + disaster recovery

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Native RDP

MOST POPULAR LOCAL MACHINE

Source machine

Source machine OS

Windows

Source IP address

Local IP | 102.33.100.237 [Connecting over a VPN?](#)

Destination VM

VM IP address

Public IP | 102.37.133.255

VM port

3389

Connection prerequisites

VM access

Port 3389 is accessible from source IP(s) [View applied NSG rules](#)

Check access

Connect using RDP file

Download and open file to connect

Download RDP file

Username

azureuser

Forgot password? [Reset password](#)

VM-Web2 | Connect

Virtual machine

Search

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- Access control (IAM)
- Tags
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Native RDP

MOST POPULAR LOCAL MACHINE

Source machine

Source machine OS Windows

Source IP address Local IP | 102.33.100.237 [Connecting over a VPN?](#)

Destination VM

VM IP address Public IP | 102.37.132.122

VM port 3389

Connection prerequisites

VM access Port 3389 is accessible from source IP(s) [View applied NSG rules](#)

[Check access](#)

Connect using RDP file

Download and open file to connect

[Download RDP file](#)

Username

azureuser

Forgot password? [Reset password](#)

VM-Web2 | Connect

Virtual machine

Search

- Overview
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Native RDP

MOST POPULAR LOCAL MACHINE

Source machine

Source machine OS Windows

Source IP address Local IP | 102.33.100.237 [Connecting over a VPN?](#)

Destination VM

VM IP address Public IP | 102.37.132.122

VM port 3389

Connection prerequisites

VM access Port 3389 is accessible from source IP(s) [View applied NSG rules](#)

[Check access](#)

Connect using RDP file

Download and open file to connect

[Download RDP file](#)

Username

azureuser

Forgot password? [Reset password](#)

Task 3: Create Load Balancer

Creating a load balancer in Azure is how you distribute traffic across multiple servers or services to ensure high availability, scalability, and performance. The exact steps depend on which type of load balancer you choose (Layer 4 vs Layer 7).

← ↻ <https://portal.azure.com/#create/Microsoft.LoadBalancer-ARM>       

Microsoft Azure Copilot        hnp@almsco@gmail.com
DEFAULT DIRECTORY RESPONSE...

Home > Create a resource > Marketplace

Create load balancer

Azure load balancer is a layer 4 load balancer that distributes incoming traffic among healthy virtual machine instances. Load balancers use a hash-based distribution algorithm. By default, it uses a 5-tuple (source IP, source port, destination IP, destination port, protocol type) hash to map traffic to available servers. Load balancers can either be internet-facing where it is accessible via public IP addresses, or internal where it is only accessible from a virtual network. Azure load balancers also support Network Address Translation (NAT) to route traffic between public and private IP addresses. [Learn more](#) 

Project details

Subscription *

Resource group * [Create new](#)

Instance details

Name *

Region *

SKU * ☒ Standard (Distribute traffic to backend resources) ☐ Gateway (Direct traffic to network virtual appliances)

Type * ☒ Public ☐ Internal

Tier * ☒ Regional ☐ Global

[Review + create](#) [< Previous](#) [Next: Frontend IP configuration >](#) [Download a template for automation](#) [Give feedback](#)

Home

Create load balancer

Basics Frontend IP configuration **Backend pools** Inbound rules Outbound rules Tags [Review + create](#)

A backend pool is a collection of resources to which your load balancer can send traffic. A backend pool can contain virtual machines, virtual machine scale sets, and containers.

[+ Add a backend pool](#)

Name	Virtual network	Resource Name	Network interface	IP address	Availability zone	Admin state
Add a backend pool to get started						

[Review + create](#) [< Previous](#) [Next: Inbound rules >](#) [Download a template for automation](#) [Give feedback](#)

Create load balancer



instances are eligible to receive traffic.

+ Add a load balancing rule

Name ↑↓	Frontend IP configuration ↑↓	Backend pool ↑↓	Health probe ↑↓	Frontend Port ↑↓	Backend port ↑↓
Add a rule to get started					

Inbound NAT rule

An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.

+ Add an inbound nat rule

Name ↑↓	Frontend IP configuration ↑↓	Service ↑↓	Target ↑↓	Frontend Port ↑↓
Add a rule to get started				

Review + create < Previous Next : Outbound rule > Download a template for automation Give feedback

Home

Create load balancer

instances are eligible to receive traffic.

+ Add a load balancing rule

Name ↑↓	Frontend IP configuration ↑↓	Backend pool ↑↓	Health probe ↑↓
Add a rule to get started			

Inbound NAT rule

An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.

+ Add an inbound nat rule

Name ↑↓	Frontend IP configuration ↑↓	Service ↑↓
Add a rule to get started		

Review + create < Previous Next : Outbound rule > Download a template for automation Give feedback

Add inbound NAT rule

LB-Web

Frontend IP address *

LB-Web-PublicIP (To be created)

Frontend Port *

80

Service Tag *

Custom

Backend port *

80

Protocol

TCP

UDP

Enable TCP Reset

Idle timeout (minutes) *

4

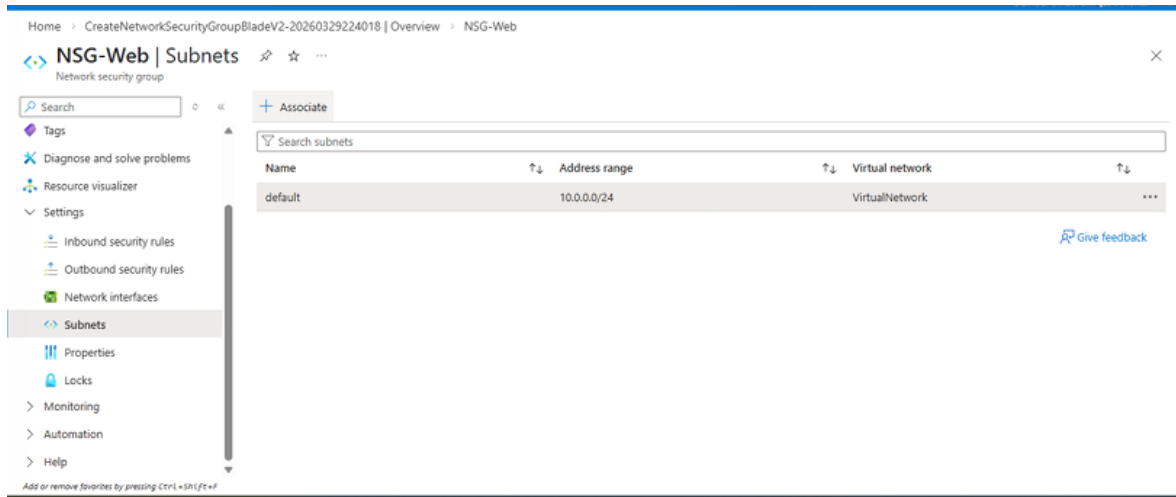
Enable Floating IP

Save

Cancel

Give feedback

Task 4: Apply Network Security Group (NSG)



A **Network Security Group (NSG)** in Azure is used to control inbound and outbound traffic to resources within a Virtual Network, such as virtual machines, subnets, or network interfaces. To configure the NSG, the basic settings were completed by selecting the same subscription, specifying the resource group, assigning an NSG name, and choosing the appropriate region for the server. Security rules were then added to enhance network protection, including a rule to block Remote Desktop Protocol (RDP) access from the internet by setting the priority to 100, the source to any, the destination port to 3389, and the action to deny. This ensures that unauthorized external access is restricted, allowing only required ports to remain open for secure communication. After completing the configuration and testing, the resource group can be deleted to reduce resource costs and improve overall efficiency in the network.